① A modeler is building a model for the call center of Student Affairs Development in the varidation phase, output results of the simulation model will be compared with the observed system behaviour which has a mean number of 73 calls per day 6 independent replications of the model were run and the following results are obtained as the average number of calls.

$S.E = \frac{s}{\sqrt{n}}$

Replication r	1	2	3	4	5	6
Sample mean for replication r	73	66	63	60	64	70

a) Develop and conduct a statistical test to evaluate whether model output is consistent .. system behaviour for the significance levels $\boxed{\alpha = 0.10}$ → two tail. one tail:
α: 0.05 and α: 0,01.

$H_0 \Rightarrow$ mean = 73
$H_1 \Rightarrow$ mean ≠ 73

$\cancel{\bar{y} \pm \sqrt{\frac{\alpha}{2}} \cdot (n-1) \cdot \frac{s}{\sqrt{n}}}$

$\bar{y} \pm t_{\alpha/2, n-1} \; s/\sqrt{n}$

$\bar{y}$ $\bar{x}$ $\mu = \frac{73+66+63+60+64+70}{6} = 66$

sample mean

variance $= s^2 = \frac{49+0^2+9+36+4+16}{5} = 22,8 \Rightarrow s = 4,77$ → standard deviation

$\bar{y} \pm \sqrt{\alpha/2} \cdot n-1 \cdot \frac{s}{\sqrt{n}}$

$|t_0| = \left|\frac{\bar{y}_{2} - \mu_0}{s/\sqrt{n}}\right| = \left|\frac{66-73}{4,77/\sqrt{6}}\right| = \frac{6}{1,95} \boxed{= 3,08}$

For $\alpha = 0.10$ (in one tail, 0,05) = 2,015 → rejected

(0,025) $\alpha = 0,05$ = 2,571

(0,005) $\alpha = 0,01$ = 4,032 → not rejected.

confidence

$$\boxed{\bar{y} \pm t_{\alpha, n-1} \cdot \frac{s}{\sqrt{n}}}$$

$3,08 > 2,015$
$2,571$

$66 \mp 2,015 \cdot 1,95$
$3,6$
$3,05$

$(62,08 , 69,92)$ $62,08 < - < 69,92$
confidence interval

$|t_0| = \left|\frac{\bar{y}_2 - \mu_0}{s/\sqrt{n}}\right|$

b) What is the power of this test if the 5 calls is viewed as critical (at most 5 calls) for the significance level $a = 0.05$?

$\delta = \frac{|E(Y) - \mu|}{s} = \frac{5}{4,77} = 1,05$ $\beta = 0,75$ $(1-0,75) = 0,25$

c) What sample size is needed to have a power of 0.9 or (critical) for the significance level $a = 0.05$?

$\beta = (1-0,9) = 0,1$ (12-13)

$$f = n \cdot \frac{e^{-a} \cdot a^{x}}{x!}$$

② Asume that the model defined in Problem 1 has been validated. Now, the Modeler is testing a system performance for new system designs. For a new design, the simulation analyst made $R = 10$ replication. Sample mean of number of calls (succesfully replied) for each replication is given in the table below

Replication r	1	2	3	4	5	6	7	8	9	10
Sample mean for replication r	88	86	80	75	65	75	69	81	77	79

a) Estimate the long run avarage number of calls and find the error of estimate by 95% and 99% Confidence intervals (find the confidence intervals.)

$\alpha = 0,05$ $\qquad$ $\alpha = 0,01$

$\alpha = 0.05$

$\alpha = 0.01$

$\bar{y} = 77,5$

$\sigma = 7,03$

$\bar{y} \pm t_{0,025}\, s/\sqrt{n} = 77,5 \pm 2,262 \times 7,03 / \sqrt{10} = 77,5 \pm 5,03$

$\bar{y} + t_{0,005}\, s/\sqrt{n} = 77,5 \pm 3,250 \times 7,03 / \sqrt{10} = 77,5 \pm 7,23$

$\frac{\alpha}{2} \rightarrow$ Dre tal $\alpha = 0,05$ $\quad \alpha = 0,01$

$77,5 \pm 2,262 \cdot \frac{7,03}{\sqrt{10}} = 77,5 \pm 5,03 \rightarrow (72,\cancel{4}7 - 82,8)$

$(\qquad)$

$77,5 \pm 3,25$

③ Assume that you have developed the model of a software development company with two system designs. In the first system design there are two developers working in a team and in the second system design there are three developers working in a team. When the two systems are simulated for 8 replications, the sample means for the time to complete a work (in terms of hours) is given below. In each system design, same random numbers are used for the same events.

Find out whether the second system is better than the first one (in terms of time to complete a work) for the confidence levels 98% and 99%?

$\alpha = 0,02$

$\alpha = 0,01$

Rep. #	System 1	System 2	difference
1	205	209	−4
2	210	215	−5
3	208	202	6
4	199	203	−4
5	207	193	14
6	211	201	10
7	206	201	5
8	205	199	6
mean →	206,375	202,875	3,5
s^2 →	13,696	43,554	50,286

No evidence that one system is better than another.

eğer 0'sa dahilse karşılaştıramayız.

tamamı 0'ın solundaysa ikinci sistem daha iyi

tamamı 0'ın sağındaysa birinci sistem daha iyi

Standard error: $\frac{s}{\sqrt{n}} = \frac{\sqrt{50,286}}{\sqrt{8}} = 2,507$

$\alpha = 0,02$

$\alpha = 0,01$

$\bar{y} \pm t \cdot \frac{s}{\sqrt{n}}$

$3,5 \pm \underbrace{(2,998) \cdot (2,507)}_{7,516} = (-4 ; 11) \rightarrow \times \; (2,11)$

$3,5 \pm \underbrace{(3,499) \cdot (2,507)}_{8,7} = (-5 , 12) \; \times$

$(-5, -2)$

Do not reject | Correct decision | Type II error |

④ A simulation will be conducted for a car washing facility where various car models will be included in the simulation. The facility manager says that service time is related with the .. of the car. (It means that in the simulation, car size and service time might not be generated separately from different distribution). Therefore, data were collected for the next 10 cars.

Car size (Length in meters)	4.65	4.36	4.78	4.88	4,80	4.42	4.88	4.9	4.35	4.07	4,69
washing Time (minutes)	6.1	5.3	7.3	8.2	7.4	5.8	8.4	9.1	5.1	4.6	6,73

Compute the sample correlation between the car size and the washing time and find out if the long time is dependent to car size?

$$\hat{cov} = [\underline{314{,}028} - (\overset{n}{10})\,(\overset{c.s.}{4.609})\,(\overset{w.t.}{6{,}721})] / (10-1)$$

$$(314{,}028 - 309{,}77)/9 = \underline{0{,}47}$$

Covariance.

$$\sum_{j=1}^{10} x_j\, x_{2j} = 314{,}028$$

$$\rho = \frac{cov}{\sigma_1 . \sigma_2} = \frac{0{,}47}{(0{,}29)(1{,}56)} = \boxed{1{,}04}$$

→ correlation. bağ güçlü

bağ ↓

0,1

$$corr = \frac{\hat{cov}}{\sigma_1 . \sigma_2}$$

$$\rho = \frac{\hat{cov}}{\sigma_1 . \sigma_2} = 0$$

1, −1 . yakınsa bağ güçlüdür.

0 'yakınsa bağ zayıftır.

0 'sa. bağ yoktur.

→ The power may also be expressed as 1 minus the probability of a Type II. or β. $\beta = P(\text{Type II error}) = P(\text{failing to reject } H_0 / H_1 \text{ is true})$

$\delta = \dfrac{|E(Y_2) - \mu_0|}{\sigma}$ For $\mu_0 = 4.3$ minutes, by 1 minute; $\alpha = 0,05$

$\dfrac{1}{1,66} = 0,60$ $\beta = (\hat{\delta}) = \beta(0,6) = 0,75$ for $n = 6$

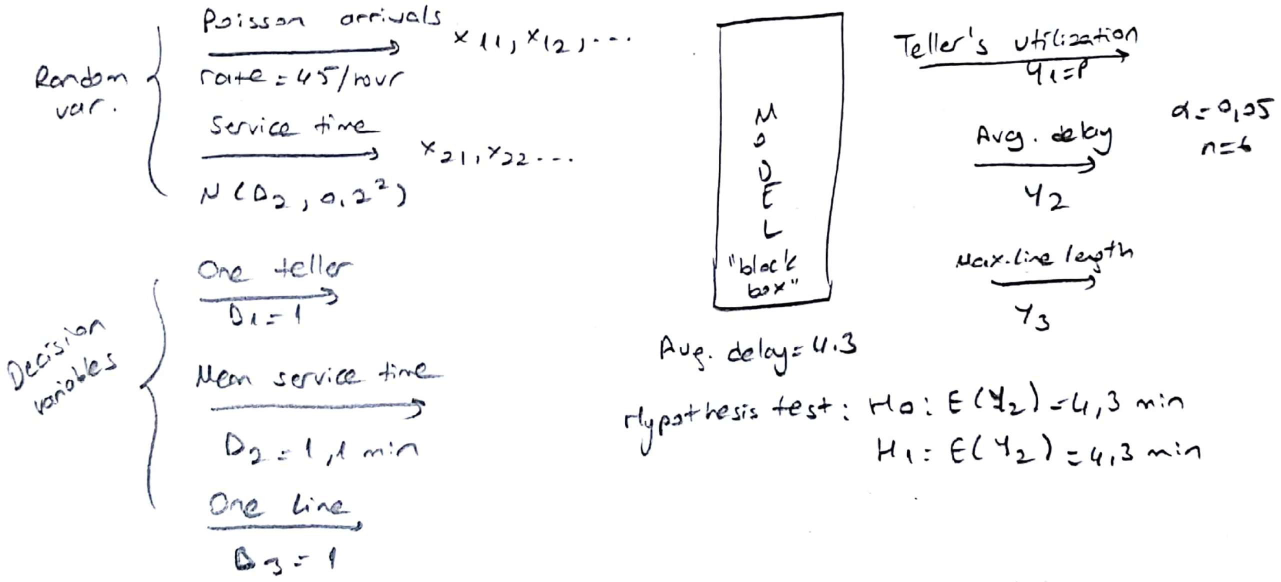
→ Verification : Building the model correctly (correctly implemented with good. input and structure) "Did we build the model right"?

Validation : Building the correct model (an accurate representation of the real system) "Did we built the right model"

Model Building:

- Observing the real system and the interactions
- Construction of conceptual model
- Implementation of an operation model

Ex



Random var.
Poisson arrivals
rate = 45/hour
$X_{11}, X_{12}, \dots$
Service time
$N(D_2, 0,2^2)$
$X_{21}, X_{22} \dots$
Decision variables
One teller
$D_1 = 1$
Mean service time
$D_2 = 1,1$ min
One line
$D_3 = 1$
MODEL
"black box"
Teller's utilization
$Y_1 = \rho$
Avg. delay
Y_2
Max. line length
Y_3

$\alpha = 0,05$
$n = 6$

Avg. delay = 4.3

Hypothesis test: $H_0 : E(Y_2) = 4,3$ min
$H_1 : E(Y_2) = 4,3$ min

$\bar{Y}_2 = \frac{1}{n} \sum_{i=1}^{n} Y_{2i} = 2,51$ min

$s = 0,82$ $t_{0,025,5} = 2,571$

$t_0 = \dfrac{\bar{Y}_2 - \mu_0}{s/\sqrt{n}} = \left| \dfrac{2,51 - 4,3}{0,82/\sqrt{6}} \right| = 5,34$

5,34 > 2,571 so Ho is rejected

Rep	Y_4	Y_5	Y_2 = Avg. Delay
1	51	1.07	2.79
2	40	1.12	1.12
3	49,5	1.06	2.24
4	50,5	1.10	3.45
5	53	1.09	3.13
6	49	1.07	2.38
Sample mean			

	when H_0 is true	when H_1 is true
Do not Reject H_0	Correct decision $p = 1 - \alpha$	Type II error $p = \beta$
Reject H_0	Type I error $p = \alpha$	correct decision $p = 1 - \beta$

Ex: Suppose that the overall avg. of the avg. cycle time on 120 rep. of a manu. sim. is 5,80 hours, std. dev. is 1.60 hours. Since $t_{0,025,119} = 1,98$ a %95 confidence interval for the long-run expected daily avg. cycle time is

$$5.80 \pm 1.98\,(1.60)\sqrt{1+\frac{1}{20}} = 5.80 \pm 3.18$$

Ex: Estimate the agent's utilization ρ over the first 2 hours of the workday.

$R_0 = 4$, $S_0^2 = 0.00518$

$\varepsilon = 0.04$

$$\left(\frac{z_{0,025}\,S_0}{\varepsilon}\right)^2 = \frac{1.96^2 * 0.00518}{0.04^2} = 12.14$$

R	13	14	15
$t_{0,0025,R-1}$	2.18	2.16	2.14
$(t_{\alpha/2,R-1}\,S_0/\varepsilon)^2$	15.39	15.1	14.83
$\underbrace{t_{\alpha/2,R-1}\frac{S}{\sqrt{R}}}_{\varepsilon}$	0.04353	0.04156	0.03987

$R = 15$ is the smallest integer

$R - R_0 = 11$

Ex Suppose $R = 1000$ reps, to estimate the $p = 0.8$ quantile with a 95% confidence interval

Output	Rank
180.42	779
→ 188.96	780
190.55	781
208.58	799
212.03	800 ←
216.99	801
250.32	819
256.79	820
256.99	821

$1000 \cdot (0,8) = 800^{th}$ smallest value.

$$P_L = 0.8 - 1,96\sqrt{\frac{.8(1-.8)}{1000-1}} = 0.78$$

$$P_u = 0,8 + 1,96\sqrt{\frac{.8(1-.8)}{1000-1}} = 0.82$$

The point estimate with 95% c.i is [188.96, 256.79)